

Kalyani Sunkara
AI Engineer

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PROFESSIONAL SUMMARY:

AI Engineer with 5 years of experience designing, developing, and deploying scalable Artificial Intelligence (AI) and Machine Learning (ML) solutions, specializing in Retrieval-Augmented Generation (RAG) pipelines, Large Language Model (LLM) applications, Generative AI chatbots, and AI agents for enterprise-scale deployments. Experienced in adapting foundation models including GPT-4o, Claude, LLaMA, and BERT using prompt engineering, Retrieval-Augmented Generation (RAG), PEFT/LoRA, and inference optimization. Proficient in optimizing LLM orchestration and retrieval workflows using LangChain, AutoGen, LangGraph, vector databases, and GPT-4o-based applications with a focus on prompt engineering, inference optimization, and API cost efficiency. Hands-on experience designing, deploying, and optimizing production-grade AI applications across AWS, Azure, and GCP with emphasis on scalability, reliability, performance, and MLOps. Skilled in semantic search, embedding pipelines, vector databases, retrieval optimization, and end-to-end AI/ML solution development. Strong background in classical machine learning, deep learning, computer vision, OCR, speech recognition, and MLOps, with expertise in regression, decision trees, random forests, XGBoost, SVMs, Docker, Kubernetes, CI/CD pipelines, and cloud-native AI deployments.

TECHNICAL SKILLS:

Programming Languages: Python, SQL (MySQL, PostgreSQL), MATLAB.

Libraries & Libraries : PyTorch, TensorFlow, Keras, OpenCV, Scikit-learn, Pandas, Hugging Face, NLTK, SpaCy, LangChain, LangGraph, RESTful APIs (FastAPI), OpenAI API, GitHub Copilot, BERT, Transformers.

AI/ML: MLOps, Generative AI, Natural Language Processing (NLP), Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Prompt Engineering, Computer Vision, Reinforcement Learning, Exploratory Data Analysis, Data Visualization, ETL, Vector Databases (Pinecone, FAISS).

Cloud: AWS- SageMaker, Lambda, S3, CodePipeline; Azure- Azure ML, Azure OpenAI; GCP - Vertex AI, BigQuery, Gemini AI, GKE

Tools: Git, GitHub, Docker, Kubernetes, Redis, Streamlit, Tableau, Spark, MS Office, Anaconda, ROS, Linux.

PROFESSIONAL EXPERIENCE:

Client: Verisk Analytics | Jersey City, NJ

Feb 2026 – Present

Role : AI Engineer

Project: Developed an AI-powered underwriting assistant for the insurance domain to help underwriters review policy documents, risk reports, and underwriting guidelines more efficiently. Leveraged Generative AI and RAG to provide relevant information and support faster decision-making.

- Built a production-grade RAG chatbot using Python, deployed on AWS, integrating SQL, vector databases (Pinecone, FAISS), and knowledge graphs for real-time, context-aware query handling.
- Developed OCR pipelines using Tesseract and AWS Textract to automate document text extraction for enterprise document processing.
- Optimized LLM response performance through prompt engineering (Chain-of-Thought, Few-Shot) and conversation memory management for faster and more consistent responses.
- Developed automated evaluation pipelines to assess LLM response quality, retrieval accuracy, and hallucination rates, improving answer consistency for enterprise users.
- Implemented LLM guardrails, input validation, logging, and unit testing to defend against prompt injection attacks and improve the reliability of AI decision workflows.
- Deployed GenAI pipelines on Azure ML and optimized inference workflows, improving system reliability and reducing latency in cloud-based deployments.
- Implemented LLM observability dashboards to monitor latency, token usage, retrieval quality, hallucination metrics, and user feedback, enabling continuous optimization of production AI systems.
- Enhanced the underwriting assistant by implementing LLaMA fine-tuning and RAG-based document retrieval to improve document analysis and provide accurate, context-aware responses.
- Optimized GPT-4o API utilization through token optimization, batching, and request deduplication to improve cost efficiency and support scalable AI workloads.
- Implemented hybrid retrieval combining semantic vector search and keyword-based retrieval to improve document relevance and answer precision.
- Designed and developed agentic AI workflows using LangGraph, LangChain, and AutoGen to orchestrate multi-step reasoning, tool execution, state management, and contextual decision-making across enterprise knowledge sources.
- Collaborated with cross-functional teams using Agile methodologies, participating in sprint planning, backlog refinement, and technical design discussions.

Client: QuinStreet | Foster City, CA

Mar 2025 - Jan 2026

Role : AI/ML Engineer

Project: Built machine learning solutions for customer analytics, predictive modeling, and recommendation systems on AWS. Developed scalable data pipelines and automated model deployment to support data-driven business decisions.

- Designed and deployed scalable machine learning models on AWS SageMaker, including training, tuning, and hosting ML models for classification, regression, and recommendation systems.
- Built end-to-end data pipelines for data ingestion, preprocessing, feature engineering, model training, evaluation, and deployment using AWS Glue, Lambda, Athena, S3, and SageMaker Pipelines.
- Analyzed large volumes of data from sources such as Redshift, RDS, and DynamoDB to identify patterns, build predictive models, and generate insights using Python, pandas, and SQL.
- Built NLP and computer vision workflows using AWS AI services to automate document understanding and image classification for unstructured datasets.
- Implemented CI/CD pipelines for ML model deployment using CodePipeline, CodeBuild, and SageMaker Model Registry, ensuring reproducibility and version control of ML artifacts.
- Used CloudWatch for monitoring model endpoints and implemented auto-scaling policies for performance optimization and cost control.
- Developed Retrieval-Augmented Generation (RAG) prototypes using Amazon Bedrock and vector search to evaluate enterprise document question-answering capabilities.
- Automated model retraining and deployment workflows, reducing model release cycles from days to hours while improving reproducibility across environments.
- Developed distributed machine learning pipelines using Apache Spark, MLlib, PyTorch, and Scikit-learn to train large-scale classification, regression, and clustering models for customer analytics and recommendation use cases.
- Built deep learning models using CNN architectures and Spatial Transformer Networks for image classification and object recognition, improving prediction accuracy on complex visual datasets.
- Performed data cleaning, preprocessing, feature engineering, and feature scaling using Pandas, NumPy, and Scikit-learn to improve model performance and training efficiency.
- Optimized machine learning models through hyperparameter tuning, cross-validation, feature selection, and regularization techniques, improving model accuracy while reducing overfitting.

Client: Divimaxim Technologies | India

Jun 2020 – Nov 2023

Role : Data Scientist

Project: Developed machine learning and NLP solutions for customer analytics, sentiment analysis, and predictive modeling. Built data processing pipelines and deployed ML models to support business reporting and real-time analytics.

- Designed and fine-tuned sentiment analysis models using BERT and Hugging Face Transformers to classify multilingual customer feedback for enterprise analytics applications.
- Built and deployed REST APIs using FastAPI, Docker, and AWS Elastic Beanstalk to expose machine learning services for real-time client applications.
- Deployed scalable machine learning models on AWS SageMaker and EC2 instances for production inference, setting up endpoint monitoring and performance alerts using Amazon CloudWatch.
- Analyzed customer behavior data and built predictive models using TensorFlow, Scikit-Learn, and Keras to uncover patterns in churn, sales forecasting, and risk analysis.
- Applied Decision Trees, Random Forests, and XGBoost models to develop customer retention and churn prediction solutions for e-commerce clients.
- Developed automated ETL pipelines for data ingestion, preprocessing, and feature engineering using Pandas, Scikit-Learn Pipelines, and AWS Glue.
- Conducted A/B testing and data validation to refine model accuracy and improve decision-making pipelines.
- Collaborated with business analysts and engineering teams to translate analytical findings into actionable recommendations.
- Automated model retraining workflows by integrating AWS Glue for data refresh, SageMaker Pipelines for model retraining, and AWS Lambda triggers.
- Participated in Agile ceremonies using JIRA, collaborating with cross-functional teams to deliver production-ready machine learning solutions.

EDUCATION:

- Master of Science in Computer Science | Lewis University, Romeoville, IL, USA | Jan 2024 – Dec 2025 | GPA: 3.7/4.0
- Bachelor of Engineering in Computer Science | Bhimavaram Institute of Engineering and Technology (Affiliated to JNTUK), India | June 2017 – July 2021 | GPA: 7.7/10